

Discussion: How to catch a star?

Reliability of filtering estimates in linear state space systems

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Overview

- How well identified is my model?
 - Always a good question in structural work
- Authors develop nice measures for evaluating identification in state-space models
- Reliability metrics for hidden state estimates:
 - Correlation/ R^2 measure
 - Half-life of estimation errors
- Apply to natural rate (r^*) estimation

Strengths

- Answers an important question in these models
 - How well does the filter recover true states?
- Tractable, easy for users to report (especially the R^2 measure) and intuitive!
- Accounts for parameter uncertainty in reliability measures (crucial improvement)
- Pushes the identification qualification further with a decomposition: levels vs. changes in hidden states

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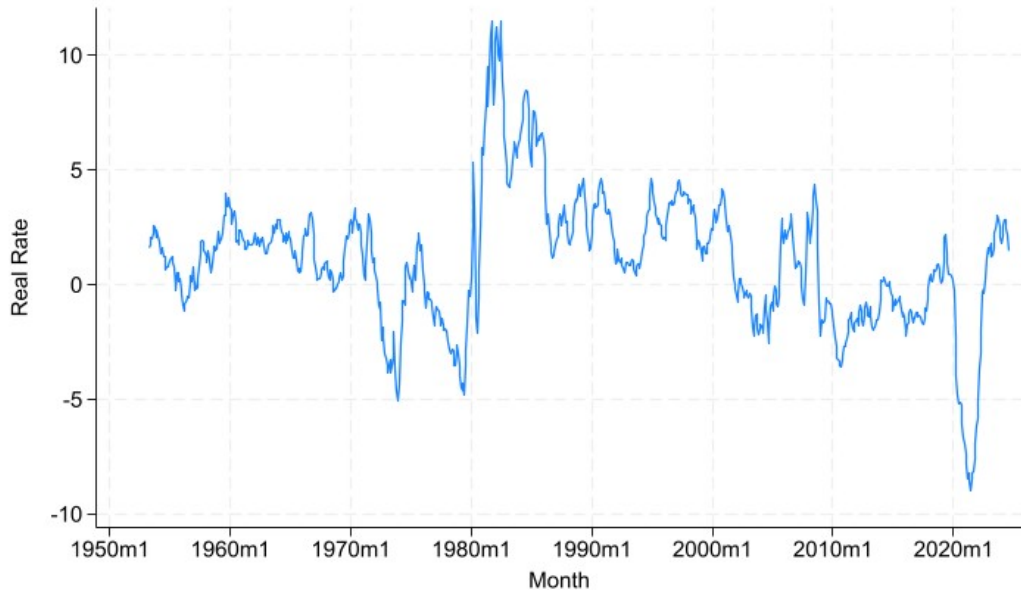
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 - Bad uses: policy advisors make arguments from well-estimated differences when their objective function is about the level
 - Good uses: maybe it can inform what types of structure or observation equations should go in these models?



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 - Real interest rates likely stationary in reality
 - Would differences still be better identified?

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 - My guess: no.
- ... but if they are, it would be helpful to define a theoretical framework that justifies comparing these models directly.

Additional Suggestions

- For take-up: can you derive additional statistical properties of these measures?
- Consider a more distinctive name for the R^2 metric
- It wasn't always crystal clear that you account for parameter uncertainty
- I don't fully understand the connection to “recoverability” (Chahrour and Jurado)

Conclusion

- Nice!
- Packaged well, this will be used.
- ... what is r^* ??